

# FALCON's Response to RFQ Weight base rejection system 2.5K

## Option-2

December 20, 2023



[www.falconautotech.com](http://www.falconautotech.com)



**Kind Attention –**

**Mr. Sandesh Mulik**

**M/s Flipkart**

**Offer Ref: F23-00535\_OP2; Date: 20-12-2023**

**Subject – Techno -Commercial Offer of Weight base rejection system 2.5K PPH**

Dear Team,

We are pleased to submit our Techno-commercial Offer in response to your RFP

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Automation Technologies.

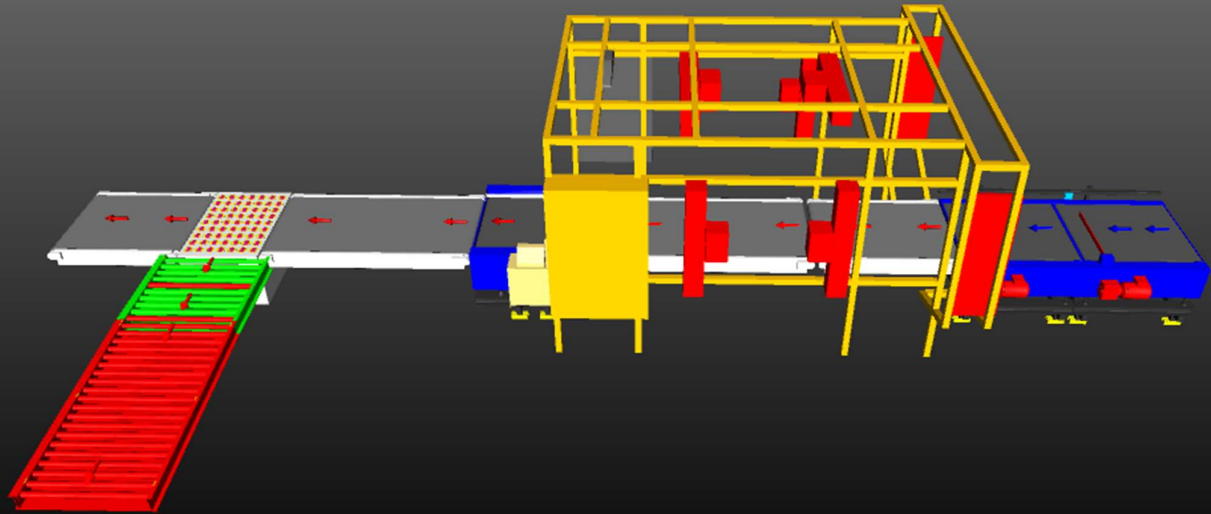
To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

**Best Regards,**

**Sandeep Bansal**

**Chief Business Officer**

## Response to RFP for Weight base rejection system 2.5K PPH Throughput



**Proposal Reference – F23-00535-OP2**

**Falcon Autotech Private Limited  
Plot No. 87, Sector Ecotech-1,  
Extention-1, Greater Noida, Uttar  
Pradesh 201308.**

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## 1. Glossary

| S. No. | Term | Description                |
|--------|------|----------------------------|
| 1      | RFP  | Request For Proposal       |
| 2      | PPH  | Parcels Per Hour           |
| 3      | IT   | Information Technology     |
| 4      | BOQ  | Bill Of Quantity           |
| 5      | I/O  | Input/ Output              |
| 6      | PDP  | Power Distribution Panel   |
| 7      | PC   | Personal Computer          |
| 8      | UPS  | Uninterrupted Power Supply |

## 2. Company Profile

Falcon Autotech (Falcon) is a global intralogistics automation solutions company. With over 10 years of experience, Falcon has worked with some of the most innovative brands in E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical Industries. With our proprietary software and robust hardware integration capabilities, Falcon designs, manufactures, supplies, implements, and maintains world-class warehouse automation systems globally. Falcon's strong research and development team and the continuous focus on innovation reflect our strong solution line around Sortation, Robotics, Conveying, Vision Systems and IOT. Falcon has done over 1,800 installations across 15 countries on four continents.

### Falcon 2.0



Operational since in 2012, Falcon is a global intra-logistic automation solution provider handling parcels, bags, cartons and pieces



Falcon has five key solution lines: 3D Robotics, Sortation systems, Dimensions & Weight Systems, Put/pick To Light & Conveyors, along with rapidly scaling proprietary software called FACTS



Work with leading E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical brands

5 Product Lines

15+ Countries with Live Installations

1800+ Total Installed Systems Globally

1Million+ Sorts Per Hour

600+ Employees

#### Long-standing Relationships with industry leaders

Ecom/ Retail

Flipkart amazon Lazada

CEP

aramex BLUE DART DELHIVERY Ecom Express

Distribution & Others

SWIGGY TITAN

#### News & Articles

**FALCON AUTOTECH EXPANDS PRESENCE IN THE MIDDLE EAST WITH NEW OFFICE OPENING**  
Falcon Autotech is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market.

**FALCON AND ALSTEF GROUP ANNOUNCE GLOBAL TECHNOLOGY PARTNERSHIP**  
Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions  
Published - Sept 2022

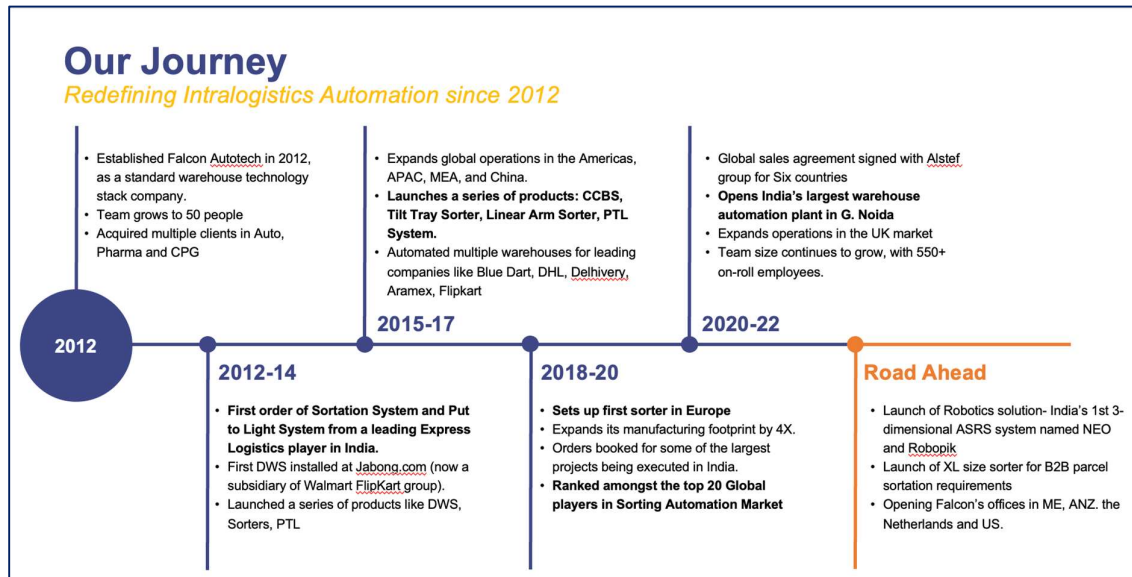
Falcon Autotech is currently among the top 15 intralogistics automation company, our vision is to become top 10 intralogistics automation company in our focused product lines.

### Our Vision

To be amongst the Top 10 global intra-logistics automation companies in our focused product lines



The team started out in 2004 solving special purpose automation problems for clients and later established Falcon Autotech in 2012 with strong focus on building standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems around warehouse automation and material handling. Over the decade, Falcon has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.



As a leading player in the intra-logistics automation space, Falcon continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions. In order to be able to live up to the high expectations set forth by our clients, the team at Falcon realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.





## Product and Solutions

With 100% focused on Parcels, Eaches, Totes, Bags, Cartons



### SORTATION SOLUTIONS

- Cross Belt Sorter
- Linear Arm Sorter
- Swivel Divert Sorter
- Tilt Tray Sorter
- PopUp Sorter
- Sweep Sorter
- Pusher Sorter



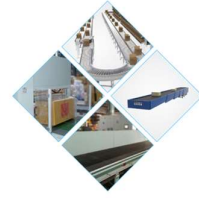
### PICK/PUT TO LIGHT SYSTEMS

- PTL Module
- Racks
- Conveyors
- Hand Scanners
- Printers
- Peripheral Displays



### DIMENSIONS & WEIGHT SCANNING SYSTEMS

- Cubizon Series
  - R, R-Eco, R-Thru, R-Cross
- Dynamic Profilers
  - Mini (600MM)
  - Jumbo (1200MM)



### CONVEYOR SOLUTIONS

- Belt Conveyors
- Roller Conveyors
- Modular Conveyors
- Special Application Solutions



### ROBOTICS

- NEO
- Robopick

Powered by

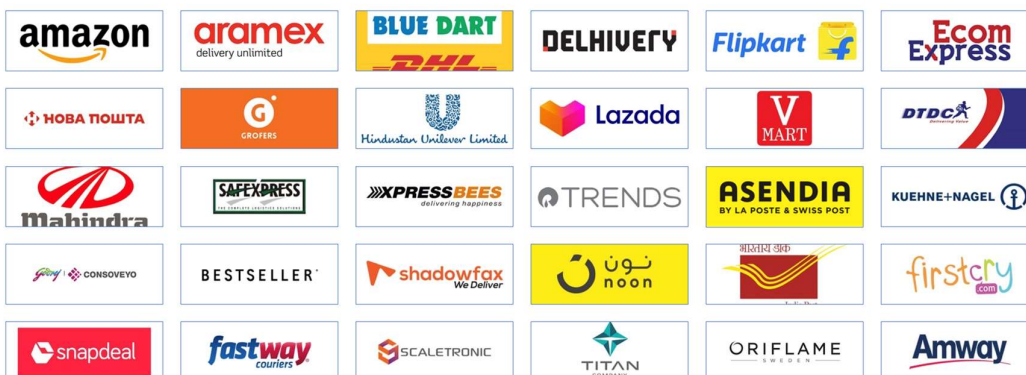


Falcon Autotech has successfully delivered warehouse automation solutions based on smart and innovative combinations of above product lines for effective materials handling, sortation and movement. The process is controlled in real-time by our in house WCS applications. These solutions considerably cut the need for manual operations, improve working conditions and ensure the highest accuracy of the entire process up to final delivery to the recipient.

Over the last 10 years, Falcon has worked with some of the most innovative brands worldwide and has established long standing partnerships. These brands are testimony of our strong focus on delivering superior customer satisfaction and offering end-to-end intralogistics solutions.

## Select Key Clients

Some of world's most innovative brands trust us with their intra-logistics warehouse automation requirements





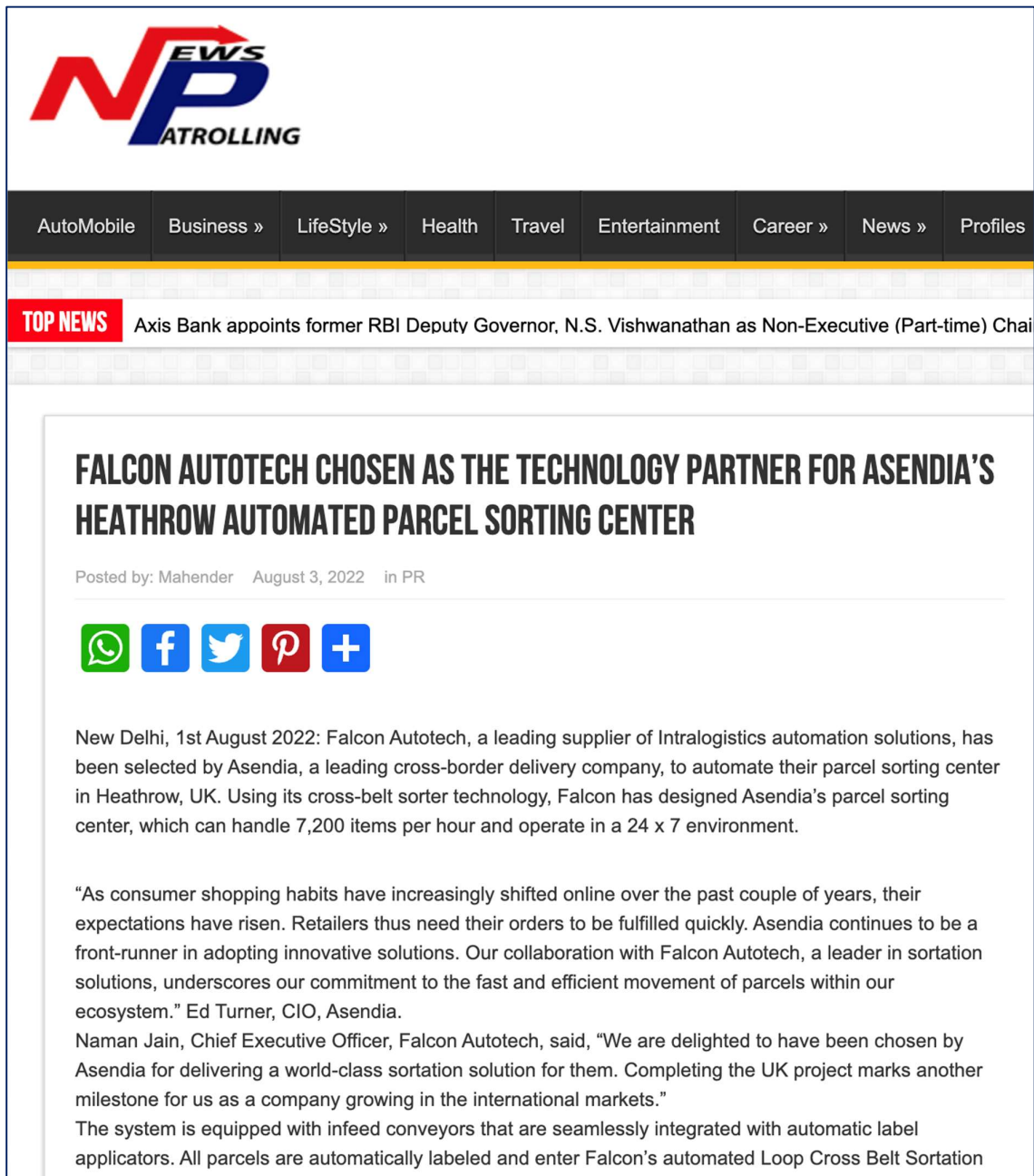
With over 1,800 installations, today Falcon's systems are used all over the globe. Falcon has highly motivated team of 600+ employees supported by over 15 global partners who help us design, manufacture, deliver and maintain automation solutions globally.

## Global Market Presence



### 3. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers globally**. Only company from APAC/EMEA Region.
- Currently possess one of the **World's largest portfolios in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 million Shipments per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.





POSTED ON APRIL 17, 2023 IN / NEWS

## Falcon Autotech Expands Presence in the Middle East with New Office Opening

Dubai, UAE – 17<sup>th</sup> April 2023 – Falcon Autotech (Falcon) is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market. The new location will enable Falcon to better serve its existing customers in the region and expand its reach to new markets and prospects.

The decision to establish a presence in the Middle East was driven by the growing demand for Falcon's solution in the region, as well as the strategic importance of the Middle East as a hub for business and innovation. With a team of experienced professionals based in the region,

## Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions

Falcon Autotech, a leading supplier of Intralogistics automation solutions, and Alstef Group, an expert in comprehensive baggage handling solutions and parcel automation integration announce a strategic technology partnership for parcel sorting solutions. As part of an exclusive distribution agreement, Alstef Group will exclusively expand the deployment of Falcon Autotech's Cross-belt sorter range to specific geographies. Falcon Autotech's Cross Belt Sorter (CBS) range includes a linear CBS and a loop CBS option. With proven features, such as automatic item centering, flexible chute positioning, wireless data communication, real-time video coding, and variable electric discharge, the Falcon CBS range is one of the fastest and smoothest sortation systems available in the market.



## 4. Reference Projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Shipment Sortation, Piece Picking and Handling, Case Picking and Handling.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focuses on Sortation Solution –

### 4.1 Project 1- (CEP Client, India)

The System is equipped with two fully automated and interconnected Sub-systems. Sub-System 1 is designed for handling large B2B boxes and E-commerce shipment bags while the Sub-System 2 is designed to handle Small E-commerce packages.

#### Solution Specifications –

- 48,000 PPH (Double Deck CBS- Shipment Sorter)
- 17,000 PPH (Double Deck CBS- Bag Sorter)
- Building Size: 700,000 Sq. Ft

#### Key Technology Modules –

- |                                                                               |                                      |
|-------------------------------------------------------------------------------|--------------------------------------|
| • 2 Sets of Double Decker CBS Sorters                                         | • Modular Belt Conveyors             |
| • Mezzanine Structures (one large mezz on to install a double decker Sorter). | • Spiral Chutes with Braking rollers |
| • Automated Singulators                                                       | • 5-Sided Scanning Tunnels           |
| • Fully Automatic Inductions                                                  | • High speed weighing conveyors      |
| • Semi-Automatic Inductions                                                   | • Direct Bagging Chutes              |
| • Telescopic belt conveyors                                                   | • Put to Light Chutes                |
| • PVC Belt Conveyors                                                          | • Volume Distribution systems        |
|                                                                               | • High Availability Server Systems   |
|                                                                               | • WCS                                |

#### Site Pictures -



#### 4.2 Project 2- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Shipments.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C shipments. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

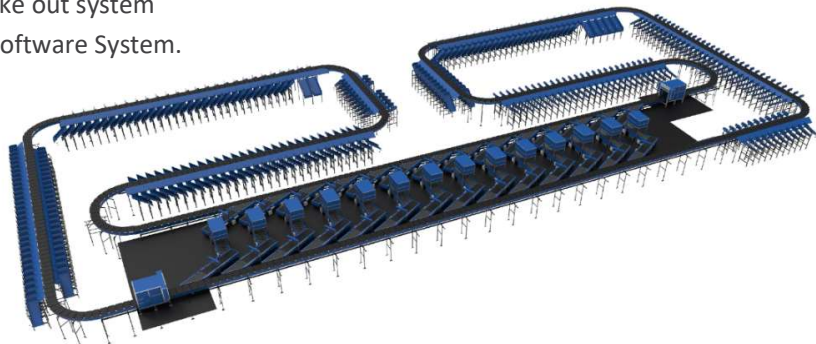
The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

##### Solution Specifications –

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq Ft

##### Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Presort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system
- WCS Software System.



### 4.3 Project 3- (Ecommerce Client, India)

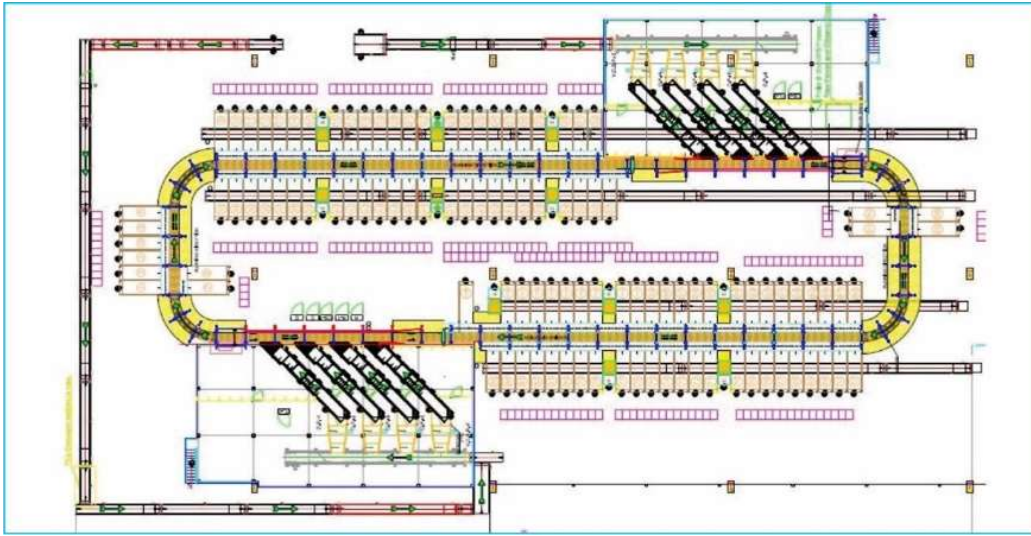
Use Case - Destination Sorting of Packed Shipments

#### Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

**Key Technology Modules** - Loop Cross Belt Sorter with a cell size of 900 x 500 mm

#### Layout and Site Pictures -





#### 4.4 Project 4- (CEP Client, UK)

This Solution is designed to handle a volume of 7200 shipments per hour. The system is equipped with three infeed conveyors integrated with an automatic label applicator before shipments enter the sortation system. The shipments are sorted using Falcon's Loop Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

##### **Solution Specifications –**

- Throughput: 7200 PPH
- End Destinations: 58 Nos

##### **Key Technology Modules –**

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- WCS Software System.

##### **Site Picture –**



#### 4.5 Project 5- (CEP Client, Sydney)

Solution is designed for handling a throughput of 16,000 shipments per hour with the help of Falcon's Loop Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the shipments at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. System is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight shipments.

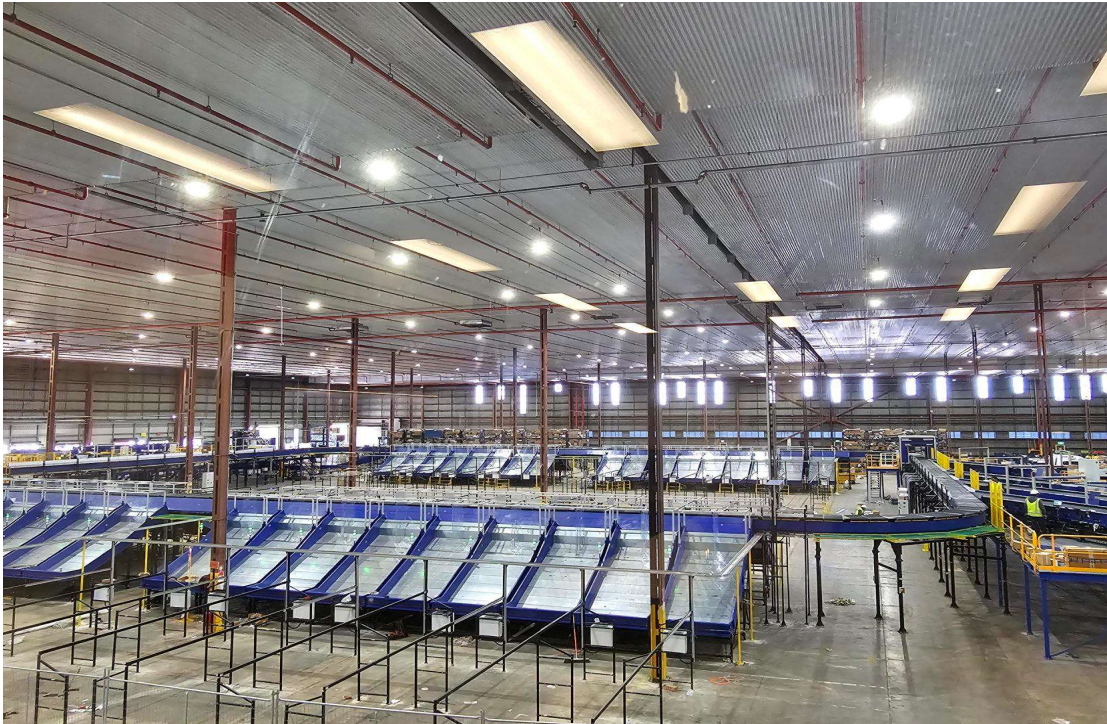
##### **Solution Specifications –**

- Throughput: 16000 PPH
- End Destinations: 369 Nos

##### **Key Technology Modules –**

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize shipment.
- Loop Cross Belt Sorter.
- WCS Software System.

##### **Site Picture –**



## 5. About Falcon

### THE COMPANY

#### ABOUT US

Falcon Autotech is one of the leading Intra-logistics automation solution providers with a sharp focus on Parcel, Piece and Case Handling applications. We have a broad portfolio of Sortation Systems for Retail, E-Commerce and CEP (Courier, Express and Postal) industries and have supplied our systems across various countries globally.

#### OUR VISION

To create a global impact on the society by facilitating world's leading organizations run their supply chains efficiently by leveraging technology enabled products and solutions and, in the process, establish Falcon Autotech as one of the world's most respected and trusted brands in Intra-logistics warehouse automation space.

#### OUR JOURNEY

The team started out in 2004 solving special purpose automation problems for clients and later pivoted to building a standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems. Over the last few years, the organization has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.

#### OUR CORE COMPETENCIES

Organizations are built and defined by the people they are able to attract, cultivate and retain and the talent management philosophy at Falcon Autotech could not agree more. At, Falcon Autotech we take a lot of pride in being the preferred career choice for some of the finest talent in the country across multiple domains such as:

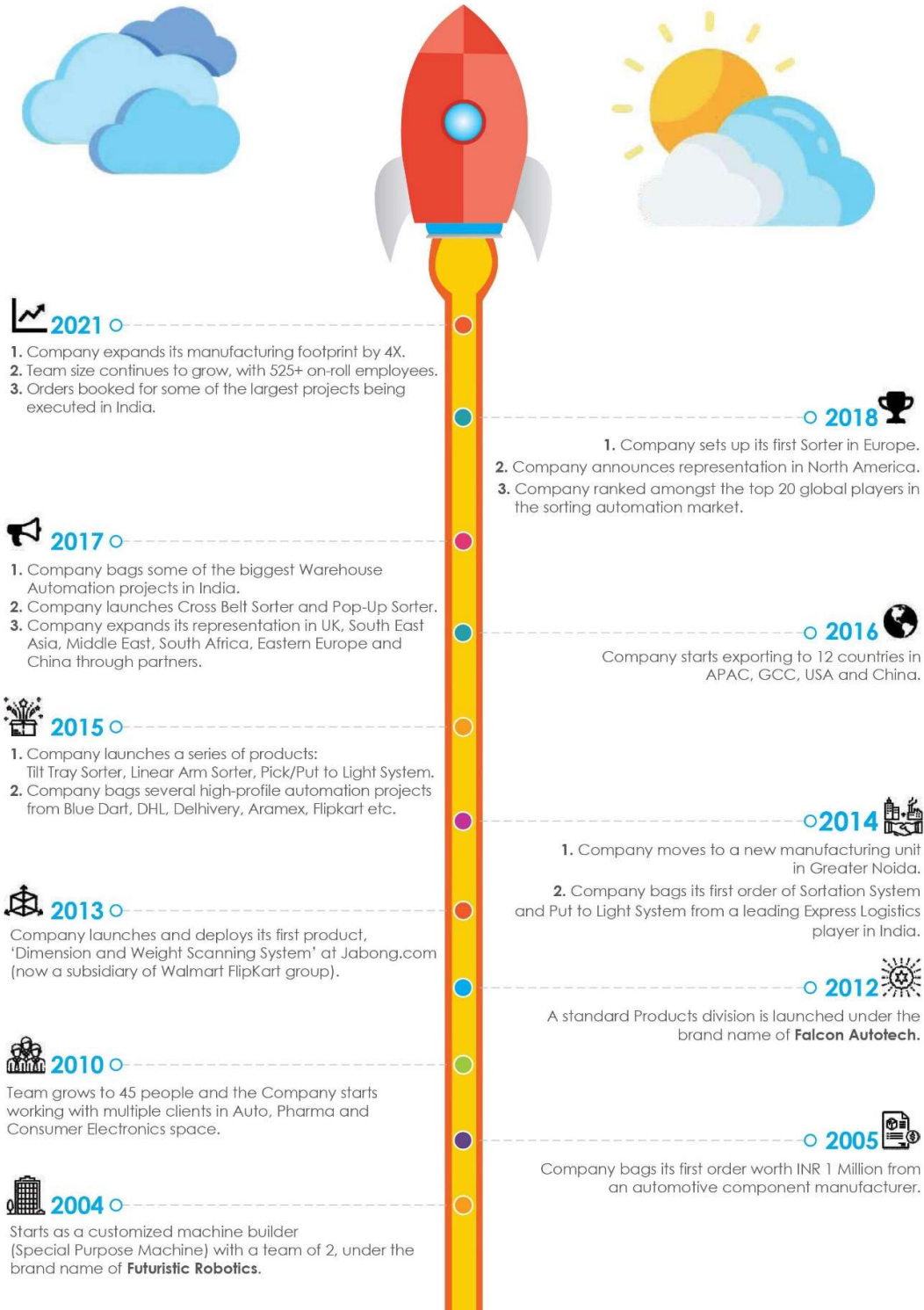
- ❖ Supply Chain Solution Designing and Consultation
- ❖ Hardware Design
- ❖ Electrical Design
- ❖ Electronics Design
- ❖ Software Development
- ❖ Manufacturing

Giving us an unprecedented ability to harness our true potential generated by culmination of Supply Chain Know-How, quality Hardware and flexible Software under one roof.





## COMPANY HISTORY



## KEY STATISTICS

---

**04**Product Lines

---

**525+**On-Roll Employees

---

**290+**Engineering Team

---

**400000+ Sq Ft**Manufacturing Space

---

**20+**Tier A Clients

---

**15+**Countries with Live Installations

---

**1800+**Total Installed Systems Globally

---

**35**Representations in 4 Continents and 35 Countries

---

**1Million+**Sorts Per Hour of combined Sorting Capacity  
installed Globally

---



## OUR PRODUCT PORTFOLIO

01

### SORTATION SOLUTIONS



Cross Belt Sorter



Tilt Tray Sorter



Arm Sorter



Swivel Wheel Sorter



Sweep Sorter



Pusher Sorter



Pop Up Sorter

02

### PICK/PUT TO LIGHT SYSTEMS



Packet Sortation



Multi-Line Order Consolidation



Order Picking



Store Order Fulfillment



Spare Parts Distribution



Kitting



Multi Order Pick Cart

03

### DIMENSION & WEIGHT SCANNING SYSTEMS

#### STATIC DIMENSION & WEIGHT SCANNING SYSTEMS



Cubizon-R



Cubizon-R THRU



Cubizon-R Cross



Cubizon-R ECO

#### DYNAMIC CONVEYOR BASED SYSTEMS

##### OUR 8 MODELS



# OUR PRODUCT PORTFOLIO

04

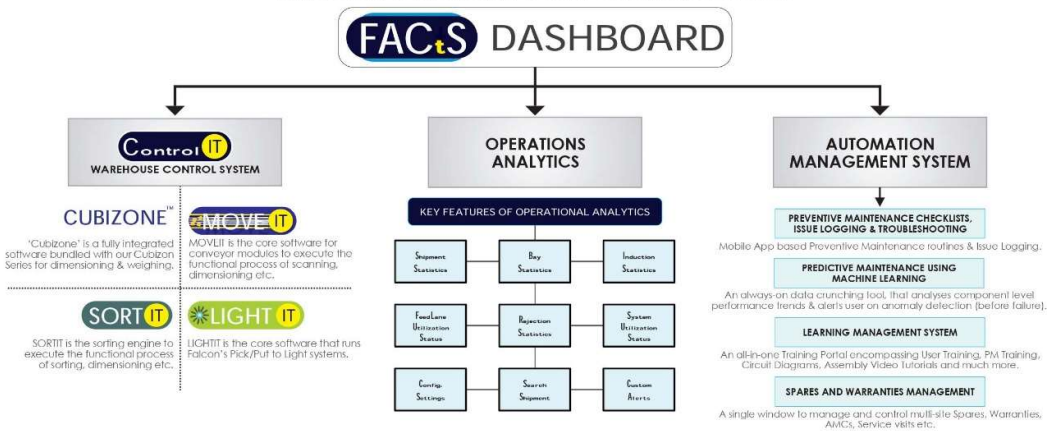
## CONVEYOR AUTOMATION SOLUTIONS

| BELT CONVEYORS                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | ROLLER CONVEYORS                                                                                                                                                                                                                                                                                                         | SPECIAL APPLICATION SOLUTIONS                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>❖ <b>PVC BELT</b> <ul style="list-style-type: none"> <li>◆ Straight Conveyor</li> <li>◆ Spacing Conveyor</li> <li>◆ Accumulation Conveyor</li> <li>◆ Strip Belt Merge- 30/45/60</li> <li>◆ Solid Belt Merge- 45</li> <li>◆ Incline Belt Conveyor</li> </ul> </li> <li>❖ <b>MODULAR BELT</b> <ul style="list-style-type: none"> <li>◆ Straight Conveyor</li> <li>◆ 90 Degree Belt Curve</li> <li>◆ 180 Degree Belt Curve</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>◆ Straight Conveyor</li> <li>◆ Gravity Roller Conveyor</li> <li>◆ MDR Accumulation</li> <li>◆ Angled Roller Conveyor</li> <li>◆ Curved Conveyor</li> <li>◆ Spur Merge Conveyor</li> <li>◆ Swivel Wheel Diverter</li> <li>◆ Pop Up Belt Diverter</li> <li>◆ Pusher Unit</li> </ul> | <ul style="list-style-type: none"> <li>◆ Aligning Conveyor</li> <li>◆ One to Two Split Conveyor</li> <li>◆ Two to One Merge Module</li> <li>◆ Bulk Distribution-ARB</li> <li>◆ Bulk Distribution-Telescopic</li> <li>◆ Bulk Distribution-Electric Arm</li> <li>◆ Telescopic Belt Conveyor</li> </ul> |

05

## SOFTWARE SUITE

### FALCON'S AUTOMATION CONTROL SUITE



### ADD-ON MODULES



#### Real Time Video Coding

No-Read Barcodes have always been a challenge in Sorter Applications globally. With Falcon's RTVC feature, we can now easily decode such challenging parcels in real time while sitting remotely.



#### Deep Learning Image Classification

DUC is an artificial intelligence function that imitates the working of the human brain. It can identify objects such as packets on the carrier and classify them in different categories.



#### Cloud Sort App/Virtual PTL

It's an android based app used to perform secondary sorting in light assisted put-away operations. It can work as a stand-alone setup or can be integrated with any sorter.



## INDUSTRY SOLUTIONS

As a leading player in the intra-logistics automation space, Falcon Autotech continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions.

In order to be able to live up to the high expectations set forth by our clients, the team at Falcon Autotech realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.

### FOCUS APPLICATIONS FOR FALCON AUTOTECH

- ❖ Piece Picking and Handling
- ❖ Case Picking and Handling
- ❖ Parcel Handling

### FOCUS INDUSTRIES



E-COMMERCE RETAIL



COURIER EXPRESS POSTAL



FASHION DISTRIBUTION



FFP (FOOD, FMCG, PHARMA) DISTRIBUTION



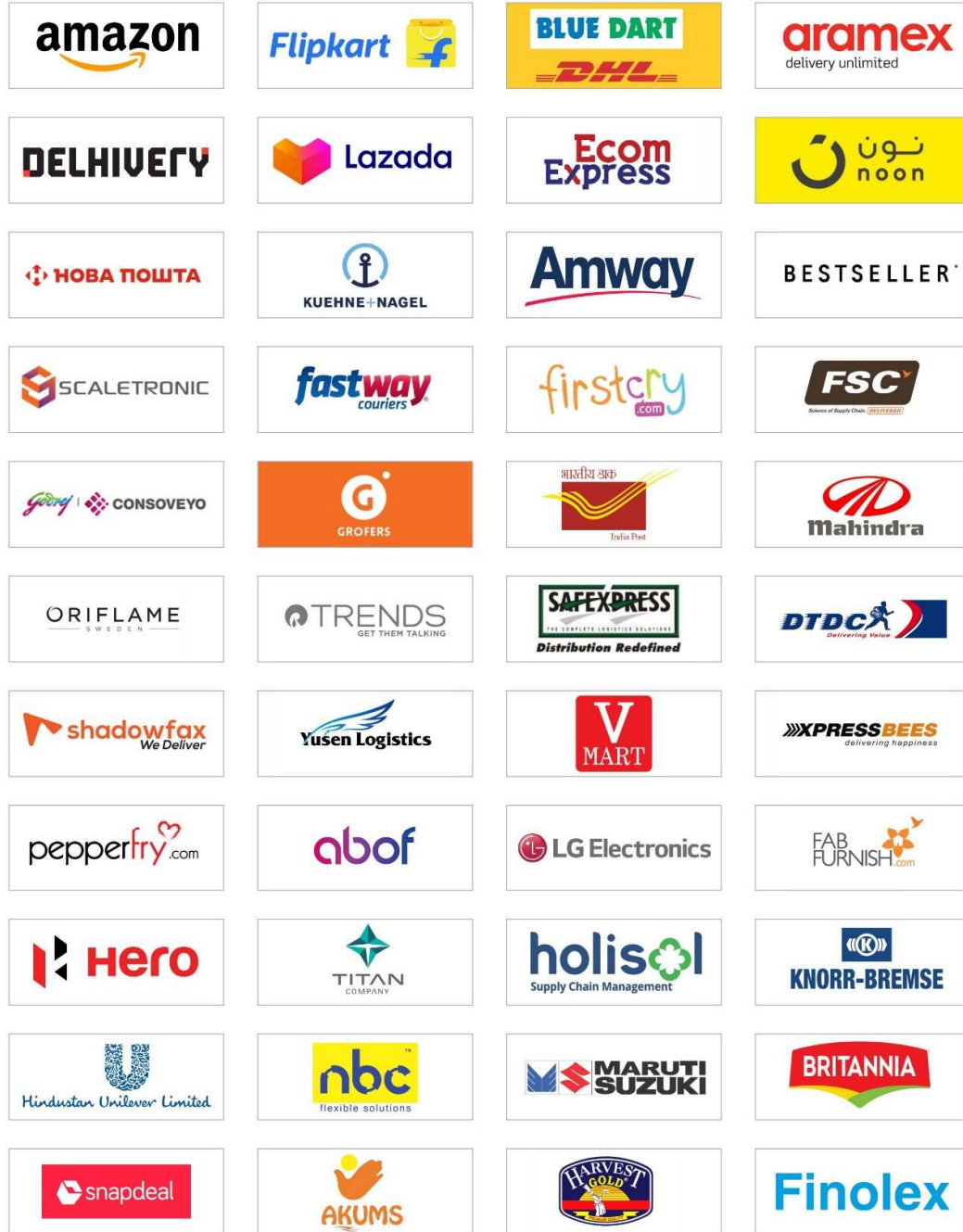
SPARE PARTS DISTRIBUTION



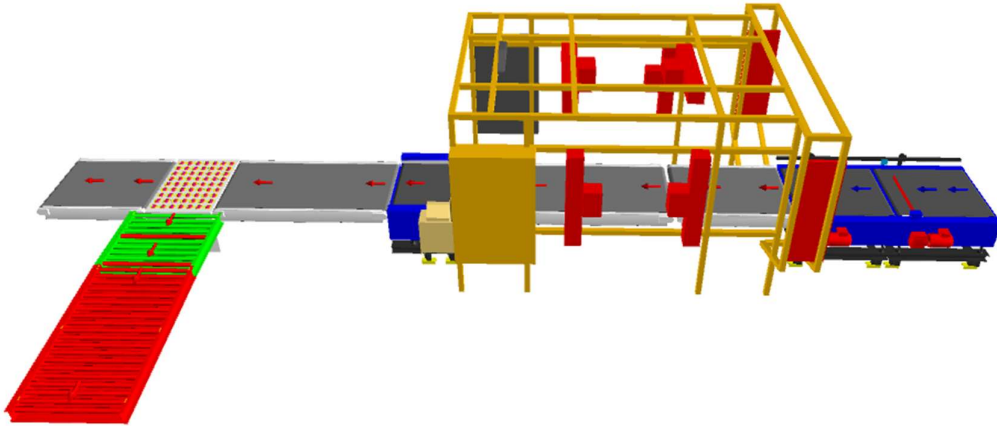
GROCERY DISTRIBUTION

## SOME OF OUR GLOBAL CLIENTS

We believe in delivering high quality and standard worldwide. Here is a gist of our esteemed clients across various industry segments.



## 6. Weight base rejection system



## 7. Handled Parcel Spectrum

### 7.1 Parcel size Conveyable on the entire system.

| Specification         | Unit | Value                                  |
|-----------------------|------|----------------------------------------|
| Max Length            | mm   | 800                                    |
| Max Width             | mm   | 800                                    |
| Max Height            | mm   | 600                                    |
| Max Weight            | Kg   | 35                                     |
| Min length            | mm   | 150                                    |
| Min Width             | mm   | 150                                    |
| Min Height            | mm   | 50                                     |
| Min Weight            | Kg   | 0.1                                    |
| Dimensioning Accuracy | Mm   | +/-10 for Length, Width and Height     |
| Weighing Accuracy     | Kg   | +/- 0.06 or 0.5 %, whichever is higher |



## 7.2 Parcels to be loaded on Conveyor/Sorter shall have the following characteristics:

1. Centre of Gravity of Item must not move during conveyance or sorting.
2. Item must not have magnetic content, otherwise behaviour of parcel cannot be guaranteed.
3. Parcels shall be perfectly and safely packed: protrusion or open surfaces are not allowed.
4. Plastic ropes shall be perfectly adherent to the surface of the package.
5. Items must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
6. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
7. Items shall not have slippery surfaces and must be able to withstand acceleration of the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
8. The parcels must have at least one flat and regular surface providing enough stability during conveyance.
9. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
10. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

## 7.3 Parcel not loadable on the system

All products that are within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll on the sorting system, such as spherical or cylindrical items
2. Items that are packed in material that can damage the conveyors or the sorter.
3. Items that have sharp points (e.g., Nails) or sharp edges, that can damage the conveyors
4. Items that have been classified as dangerous.
5. Wet items.
6. Items with anti-slip treatment.
7. Items with protruding parts.
8. Inadequately packed items that could be damaged during automatic transportation.
9. Electrostatically loaded items.
10. Loose parts on Items such as adhesive tape, stickers, slips of paper, straps, wrap foil etc.
11. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint.
12. All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system.

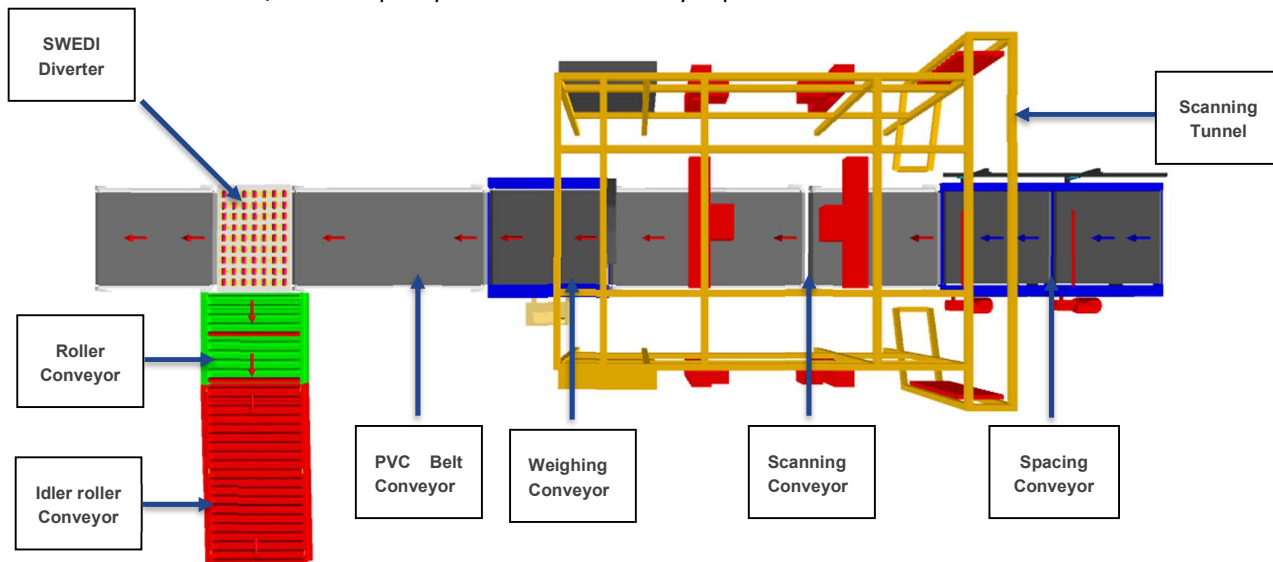
## 7.4 Types of Parcels

| Parcel Type            | Description                   |
|------------------------|-------------------------------|
| Corrugated Box/Cartons | Min: L-150 X W-150 X H-50 MM  |
| Shipments              | Max: L-800 X W-800 X H-600 MM |

## 8. Proposed System Description

### 8.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing and acceptance testing of the fully auto Weight base rejection system to handle cartons/boxes as per specifications shared by Flipkart.



**Proposed System Layout**

### 8.2 Process Flow

The shipments will be loaded on the existing conveyors placed before the spacing units. The shipment passes through spacing conveyors creating enough space between parcels and enters the automatic barcode scanning tunnel (Six Side barcode scanning) where the barcode data is read, the weight is measured and stored in the system. In case of barcode not being read or overweight the parcel will be rejected by the SWEDI divert onto the rejection roller conveyor. In case of a successful data capture & allowable weight the parcels shall head straight onto the output Conveyor.

### 8.3 Main benefits of the proposed solution

The proposed solution has the following advantages:

1. High operational throughput
2. Low occupancy of floor space in the building

## 9. System Components

### 9.1 Spacer

It is a Dual set of high-speed conveyors, using to create appropriate space between parcels/boxes.

Some Salient Features of Falcon's spacing conveyor:

- Low Noise
- Maximum uptime.
- High safety standards.
- Fastest ROI
- Comes with MS frame.



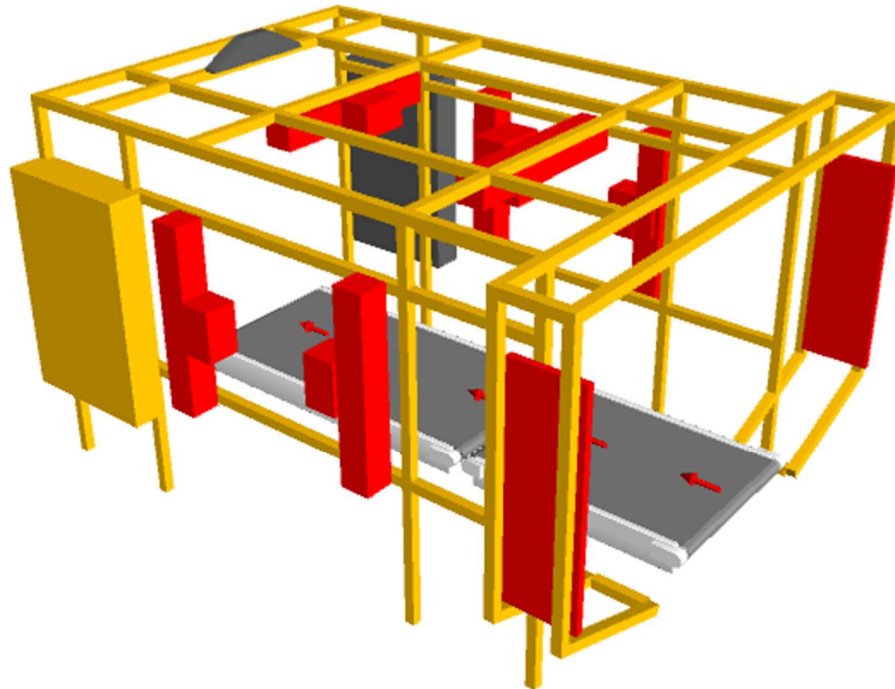
| Specification        | UOM  | Remark                                     |
|----------------------|------|--------------------------------------------|
| Manufacturer Name    | Name | Falcon Autotech                            |
| Material of Roller   | Type | MS Rollers with Zinc Plating               |
| MOC of Belt          | Type | PVC                                        |
| Belt Thickness       | mm   | 3                                          |
| Belt Finish          | Type | Matt Finish                                |
| Belt width           | mm   | 1000                                       |
| Belt Joint Type      | Type | Vulcanized, Endless Belts                  |
| Belt Make/Model No.  | Make | FORBO/DERCO                                |
| Type of Motor        | Type | AC Geared Motor                            |
| Type of Drive System | Type | Direct Shaft Mounted Motor/ Flange Mounted |
| Type of Drive        | Type | Chain drive/Torque Arm type                |

## 9.2 Auto Barcode Scanning System (Top Side)

System is integrated with automated Camera technology for scanning 1D/2D Barcodes.

Some Salient Features of Falcon's Scanning System:

- High Speed Scanning
- High Accuracy Rates
- Effortless Operation
- High safety standards.



***\*Note : Barcode data of product should be locally available for on time retrieval of weigh data against each barcode.***

***\*Note : In case of six Side scanning, skewness of more than +/- 10 deg may result in no data capture case.***



### 9.3 Weighing Conveyor

Falcon's weighing conveyor is used for inline weighing of parcels.

Some Salient Features of Falcon's weighing conveyor:

- High Speed Weighing
- Minimal maintenance
- High safety standards.
- High accuracy



| Specification       | UOM  | Remark                                                  |
|---------------------|------|---------------------------------------------------------|
| Manufacturer Name   | Name | Mettler Toledo/Bizerba                                  |
| Carton Size         | Mm   | Min: 150 x 150 x 50<br>Max: 800 x 800 x 600             |
| Minimum Weight      | Kg   | 0.1                                                     |
| Maximum Weight      | Kg   | 35                                                      |
| Weighing Accuracy   | Kg   | + - 0.06 or 0.5% of product weight, whichever is higher |
| Weighing Resolution | Kg   | 0.001                                                   |

## 9.4 Swivel Wheel Sorting Unit – Falcon

Falcon's Swivel wheel divert is used in system for the smooth transfer of products.

Some Salient Features of Falcon's Swivel Wheel Divert:

- Low Noise
- Maximum uptime.
- Electric based rotation of wheels.
- Minimal maintenance
- High safety standards.



| Specification               | UOM    | Remark                                      |
|-----------------------------|--------|---------------------------------------------|
| Manufacturer Name           | Name   | Falcon Autotech                             |
| Type - Equipment            | Type   | Merge/Sorting                               |
| Type - Operation            | Type   | Sorting                                     |
| Mechanism for Swivel        | Type   | Electric servo motor                        |
| Roller drive                | kW     | Powered smart wheel                         |
| Roller pitch (L x W)        | Type   | 110 x 120                                   |
| No. of divert sections      | Yes/No | 1 or 2                                      |
| Max. product weight (kg/m2) | kg/m2  | 25                                          |
| Throughput                  | PPH    | 2500                                        |
| Type of Mounts              | Type   | Fixed Heights Legs with Grouting Provisions |

## 9.5PVC Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

Some Salient Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



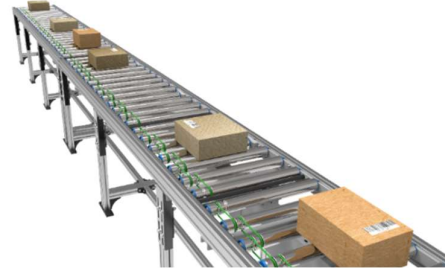
| Specification                              | UOM  | Remark                                                                                                               |
|--------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------|
| Manufacturer Name                          | Name | Falcon Autotech                                                                                                      |
| Material of Roller                         | Type | MS Rollers with Zinc Plating                                                                                         |
| MOC of Belt                                | Type | PVC                                                                                                                  |
| Belt Thickness                             | mm   | 3                                                                                                                    |
| Belt Finish                                | Type | Matt Finish                                                                                                          |
| Belt width                                 | mm   | 1000                                                                                                                 |
| Belt Joint Type                            | Type | Vulcanized, Endless Belts                                                                                            |
| Belt Make/Model No.                        | Make | FORBO/DERCO                                                                                                          |
| Chassis                                    | mm   | MS, 3mm Thick, Powder Coated-75 Microns                                                                              |
| Idler Roller Pitch on running side (Top)   | mm   | 480                                                                                                                  |
| Idler Roller Pitch on return side (Bottom) | mm   | 1200                                                                                                                 |
| Idler Roller Specs                         | Spec | Dia.50.8, shaft-12 mm, bearing-6301                                                                                  |
| End Rollers                                | Spec | Dia 81.2 x 2.85 Thick, 30mm Shaft Dia, & Bearing Size-UCFH206D1, MOC-EN9, Zinc Plated -25 microns                    |
| Drive Rollers                              | Spec | OUTER Dia 178 PIPE Dia 168x5 Thick, 40mm Shaft Dia, & Bearing Size-UCF208D1, MOC(SHAFT)-EN9, NEOPRENE RUBBER COATING |
| Tensioning Rollers                         | Spec | Dia 81.2 x 2.85 Thick, 25mm Shaft Dia, & Bearing Size-UCT205D1, MOC-MS, Zinc Plated -25 microns                      |

## 9.6 Powered Roller Conveyor

Falcon's Roller conveyors are highly energy efficient and require negligible maintenance leading to very minimal Operating costs and thus lower cost of ownership.

Some Salient Features of Falcon's roller conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI
- Comes with MS frame.



| Specification                         | UOM  | Remark                                               |
|---------------------------------------|------|------------------------------------------------------|
| Manufacturer Name                     | Name | Falcon Autotech                                      |
| Material of Roller                    | Type | MS Rollers with Zinc Plating                         |
| Overall Width                         | mm   | 1000                                                 |
| Conveyor Height                       | mm   | Variable as per design shared                        |
| Drive Power Rating                    | Watt | 40                                                   |
| Type of Motor                         | Type | DC Rollers (MDR)                                     |
| Ingress Protection                    | Type | IP 54                                                |
| Type of Drive System                  | Type | Poly V Belt Drive                                    |
| Insulation Class                      | Type | Yes                                                  |
| Conveyor Speed                        | m/s  | Variable speed to meet TPH requirements              |
| Load capacity/unit length of Conveyor | kg/m | 50                                                   |
| Roller Span                           | mm   | Variable                                             |
| Roller Diameter                       | mm   | 50                                                   |
| Pitch of Rollers                      | mm   | 75/100                                               |
| Type of Mounts                        | Type | Fixed Heights Legs with required Grouting Provisions |

## 9.7 Idler Roller Conveyor

Falcon's Roller conveyors require negligible maintenance leading to very minimal Operating costs and thus lower cost of ownership.

Some Salient Features of Falcon's idler roller conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- Comes with MS frame.



| Specification                             | UOM    | Remark                                      |
|-------------------------------------------|--------|---------------------------------------------|
| Manufacturer Name                         | Name   | Falcon Autotech                             |
| Material of Roller                        | Type   | MS Rollers with Zinc Plating                |
| Overall Width                             | mm     | 1000                                        |
| Conveyor Height                           | mm     | As per Layout requirement                   |
| Load capacity per unit length of Conveyor | kg/m   | 50                                          |
| Roller Span                               | mm     | approx. 600                                 |
| Roller Diameter                           | mm     | approx. 50                                  |
| Pitch of Rollers                          | mm     | 75                                          |
| Type of Mounts                            | Type   | Fixed Heights Legs with Grouting Provisions |
| Type of Guides                            | Type   | Yes                                         |
| End Stopper                               | yes/no | Yes                                         |



## 10. Proposed System Technical Details/BOM

### 10.1 Mechanical equipment

| S.No. | Qty. | Description                          | Value                    |
|-------|------|--------------------------------------|--------------------------|
| 1     | 1    | Weight base rejection system         |                          |
|       |      | Spacing Conveyor                     | 2 Set                    |
|       |      | Barcode Scanning Conveyor            | 2 Set                    |
|       |      | Weighing Conveyor                    | 1 Set                    |
|       |      | Automatic Barcode Scanner (Six Side) | 1 Set                    |
|       |      | PVC Belt conveyor                    | 2 Modules (Length-3.2 m) |
|       |      | SWEDI Divert unit                    | 1 Set                    |
|       |      | Powered Roller conveyor              | 1 Module (Length-1 m)    |
|       |      | Gravity Roller conveyor              | 1 Module (Length-2 m)    |

### 10.2 Electrical Equipment

| Electricals |   |                                      |          |
|-------------|---|--------------------------------------|----------|
| FC1         | 1 | <b>Electrical Cabling and Wiring</b> |          |
|             |   | Main Power Distribution panel        |          |
|             |   | Main Control Panel                   |          |
|             |   | Network Switches                     |          |
|             |   | Field Cabling                        | Included |

### 10.3 Control System

| Components |   |                                                                               |          |
|------------|---|-------------------------------------------------------------------------------|----------|
| FC2        | 1 | <b>Controls and IT Infrastructure Controls</b>                                |          |
|            |   | Weigh Scale Control, Barcode Scanner Controls,<br>Control IT Software Package | Included |

## 11. Program Organisation

### 11.1 Delivery Schedule

Dispatch within 14-15 weeks from the date of signed PO, advance and DAP whichever is later.

## 12. Commercial Terms

### 12.1 Price Sheet

| S. No             | Component                                  | Set | Price (INR)   |
|-------------------|--------------------------------------------|-----|---------------|
| 1                 | SWEDI Diverter                             | 1   | ₹ 1,21,82,633 |
| 2                 | Conveyor Automation                        | 1   | ₹ 15,05,332   |
| 3                 | Toolkit                                    | 1   | ₹ 36,000      |
| 4                 | Software Packages                          | 1   | ₹ 6,00,000    |
| 5                 | Installation & Commissioning               | 1   | ₹ 6,86,198    |
| 6                 | Packaging & Forwarding                     | 1   | ₹ 2,74,479    |
| 7                 | Project Management and Engineering Charges | 1   | ₹ 3,82,116    |
| Grand Total (INR) |                                            |     | ₹ 1,56,66,758 |

The Total Price is to be understood.

- Ex Works – Greater Noida, India
- Freight charges shall be extra at actual
- Taxes, Customs, Duties extra at actuals
- Including Packing and Forwarding
- Installation and Commissioning: Mentioned above.
- Price is valid for 60 days from the date of proposal

Our Price does not include taxes, and duties, import duties, charges, fees, taxes etc. charged by authorities outside India for deliveries and services. Same shall be borne & paid by client itself.

\*Material Unloading & Laydown area is in customer scope

### 12.2 Payment Terms

The Supplier Quotation is based on the following payment schedule:

Stage 1: 50% Advance along with PO.

Stage 2: 30% along with 100% Taxes prior to dispatch.

Stage 3: 20% Within 30 days of Commissioning.

### 13. Warranty Period

Falcons offered System comes with a standard warranty of 1 year and post completion of warranty, customer can opt for Extended warranty. Warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Faulty articles:
  - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
  - Negligence or abnormal use of equipment.
  - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
  - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

## 14.Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- Low Spec Server PC
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Server PC is not included in this proposal
- Electrical power supply and wiring to the main control cabinets
- UPS for Controls and Drives
- Electrical Power for Installation
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- Simulation and 3D animation of the sorter system
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.